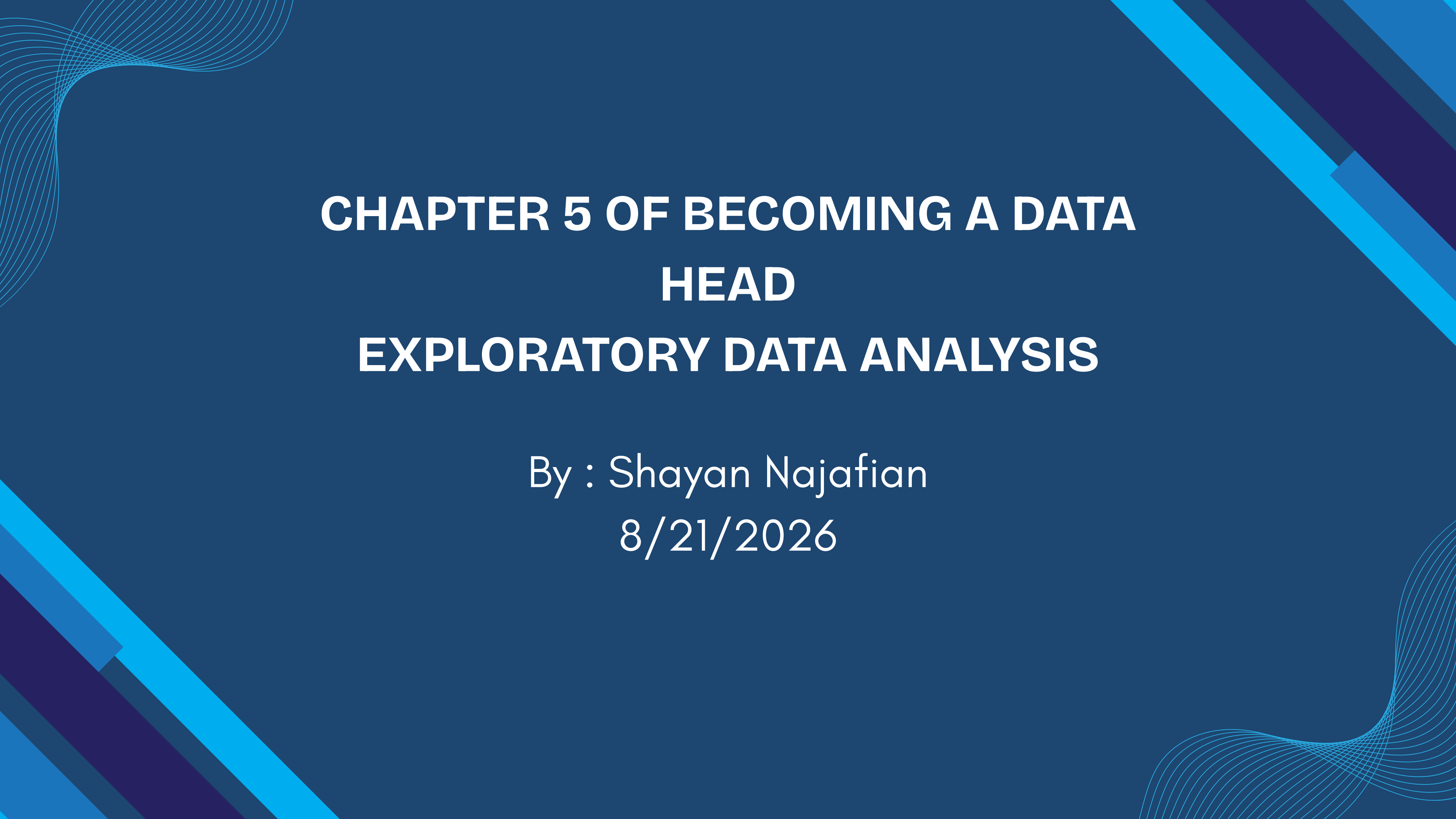




CHAPTER 5 OF BECOMING A DATA HEAD EXPLORATORY DATA ANALYSIS

By : Shayan Najafian
8/21/2026

THE CORE STORY

Explore before you model.

What is in the data?

What is unusual?

What relationships are hiding inside it?

WHAT IS EXPLORATORY DATA ANALYSIS?

EDA = Explore → Question → Learn → Repeat

Summary statistics + Visualizations

EDA IS NOT A STRAIGHT LINE

Real data analysis is iterative

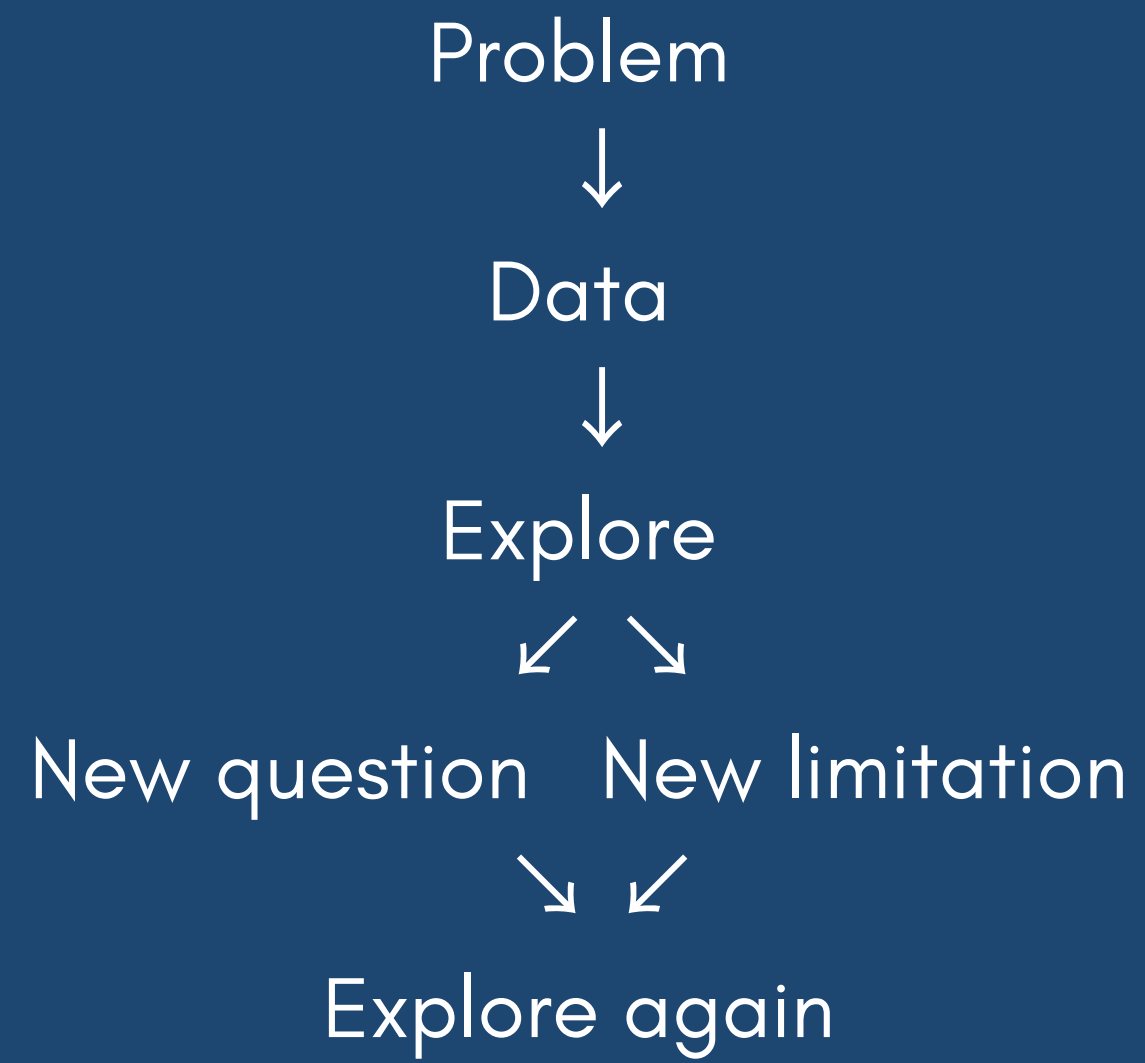
Explore → Discover → Reconsider → Explore again

Two analysts may follow different paths.

WHAT IS EXPLORATORY DATA ANALYSIS?

EDA = Explore → Question → Learn → Repeat

Summary statistics + Visualizations



WHY BUSINESS CONTEXT MATTERS

Statistically interesting $\neq$ practically useful

Data teams need:
Domain knowledge + Business context

THE THREE QUESTIONS OF EDA

1. Can the data answer the question?
2. Did you discover any relationships?
3. Did you find new opportunities in the data?

CASE STUDY: AMES HOUSING DATA

Goal: Predict home sale prices

Data: Hundreds of homes

Features: ~80 variables

Location: Ames, Iowa

Period: 2006–2011

CAN THE DATA ANSWER THE QUESTION?

Before choosing an algorithm, ask:
Does the dataset contain the information we actually need?

DO THE VALUES MAKE INTUITIVE SENSE?

Look at the actual values.
Ask:
Does this make sense in the real world?

HISTOGRAMS: SHAPE OF THE DATA

Histogram → What does the distribution look like?
House prices are right-skewed.

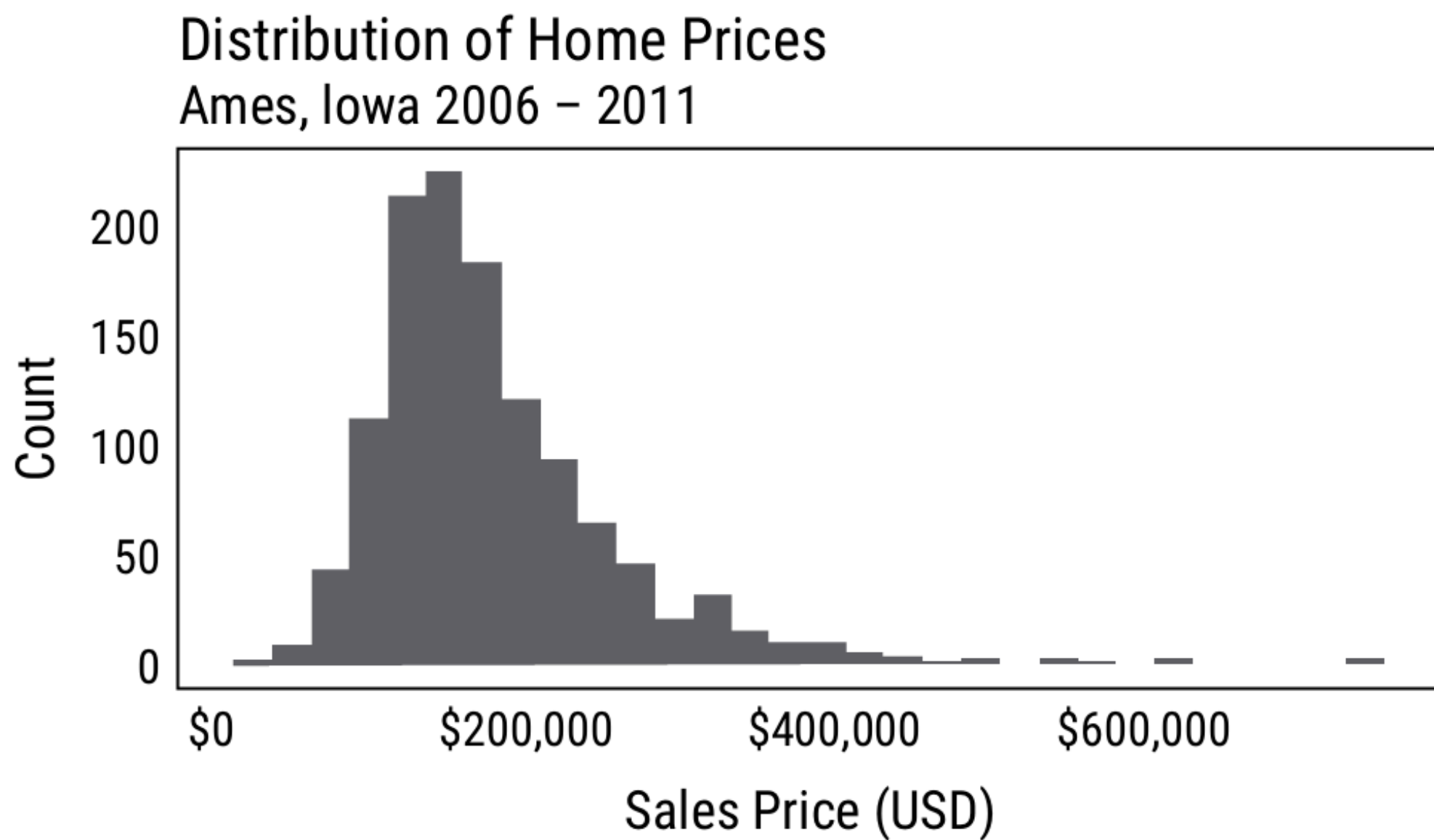


FIGURE 5.1 A histogram showing the shape of sales price

BOX PLOTS, BAR CHARTS & LINE CHARTS

Box plot → Compare groups
Bar chart → Count categories
Line chart → See trends over time

Count of Homes by Electrical Types

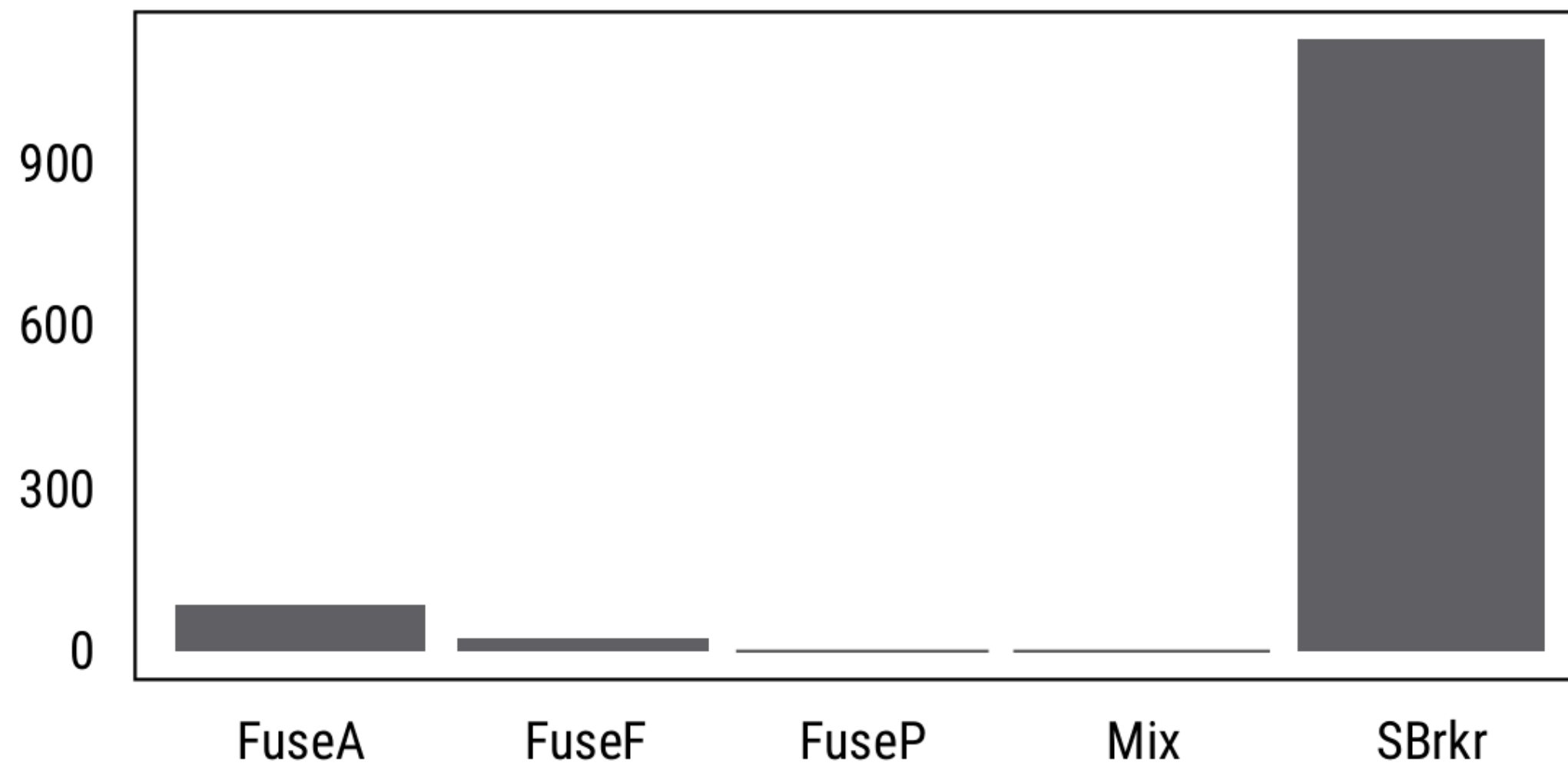
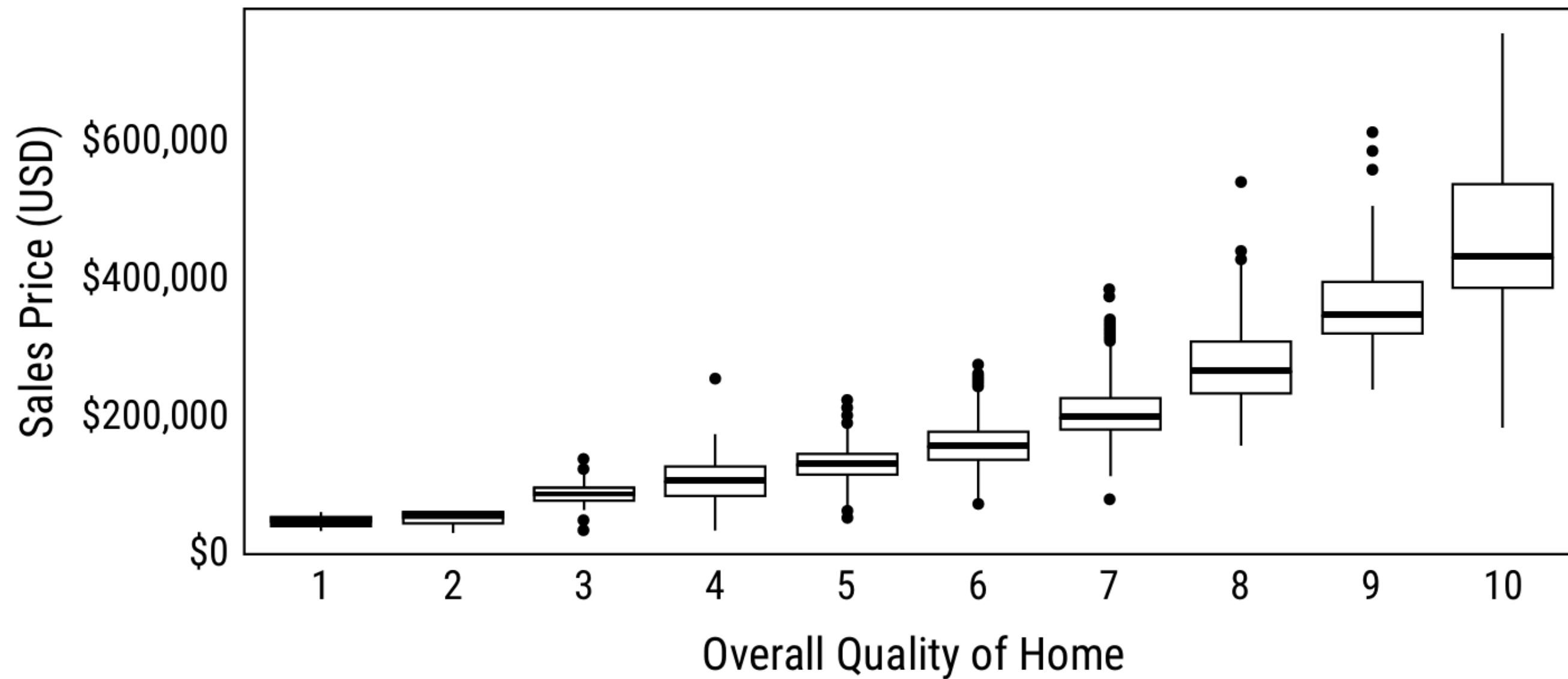
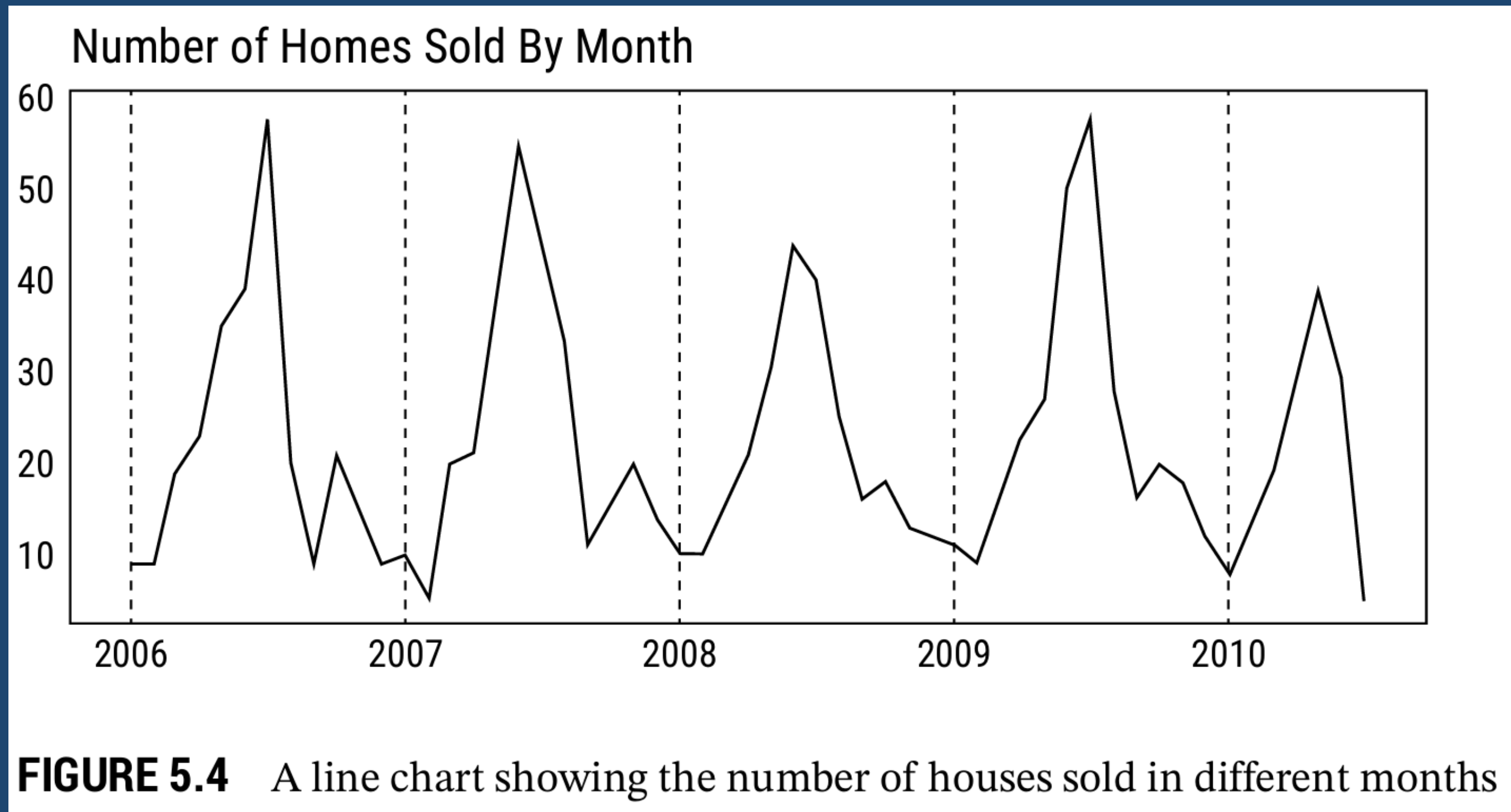


FIGURE 5.3 A bar chart showing the counts by types of electrical installation

Comparing Home Prices Ames, Iowa 2006 – 2011





SCATTER PLOTS: SEEING RELATIONSHIPS

Scatter plot → Relationship between two numeric variables
Larger homes generally sell for more.
But the relationship is not perfect.

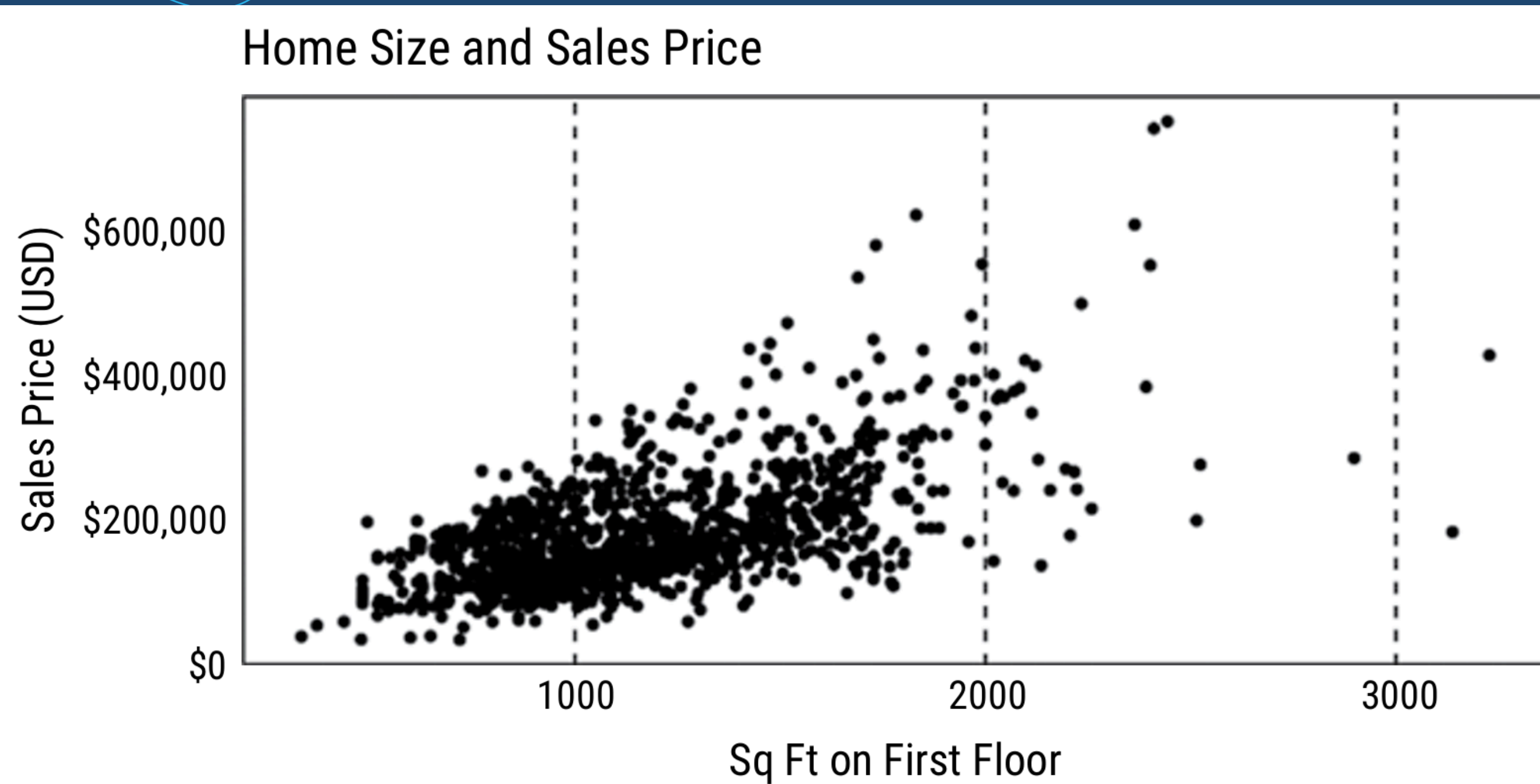


FIGURE 5.5 A scatter plot showing square footage and sales price

OUTLIERS AND MISSING VALUES

Outlier $\neq$ Error

Missing $\neq$ Zero

Context determines what they mean.

DID YOU DISCOVER ANY RELATIONSHIPS?

Relationships can help us:

- Understand the data
- Choose useful predictors
- Find redundant variables

UNDERSTANDING CORRELATION

$$-1 \leq r \leq 1$$

Positive: $X \uparrow$, Y tends to $\uparrow$

Negative: $X \uparrow$, Y tends to $\downarrow$

Near 0: little linear relationship

CORRELATION DOES NOT IMPLY CAUSATION

X moves with Y

does not automatically mean:

X causes Y

Look for confounders and alternative explanations.



HOT WEATHER

/ \



ICE CREAM SALES MORE SWIMMING



SHARK ATTACKS

'A necessary book for our times.
But also just great fun'
Saul Perlmutter, Nobel laureate



Calling Bullsh*t

The Art of
Scepticism
in a
Data-Driven
World

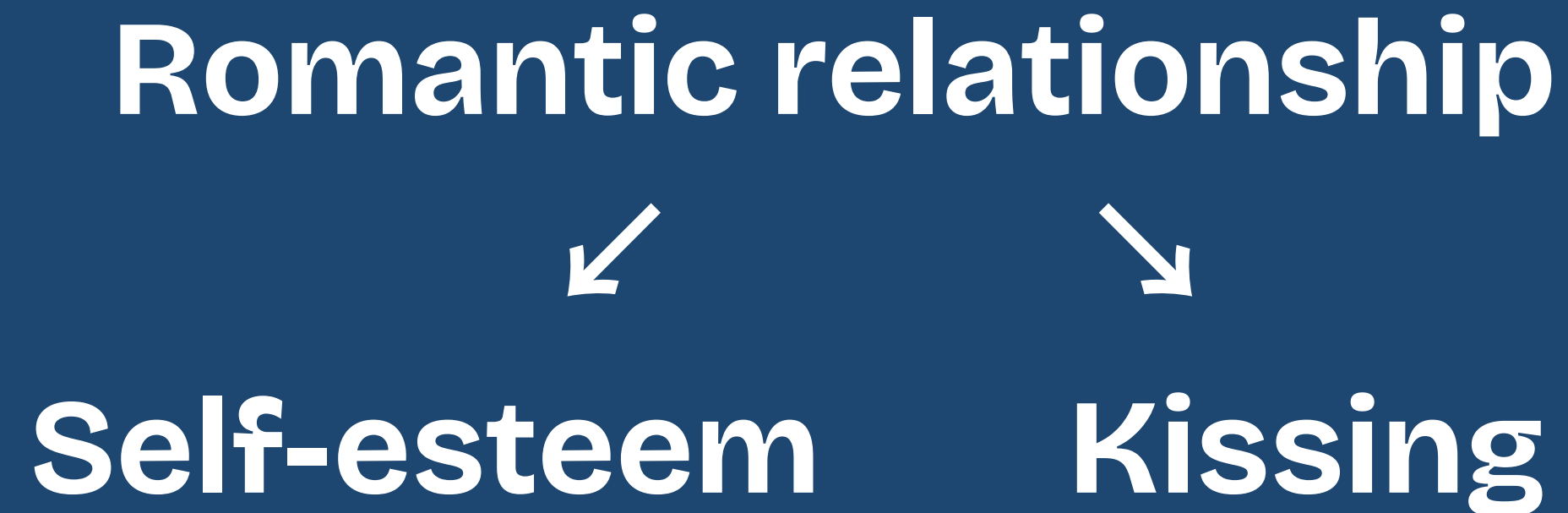


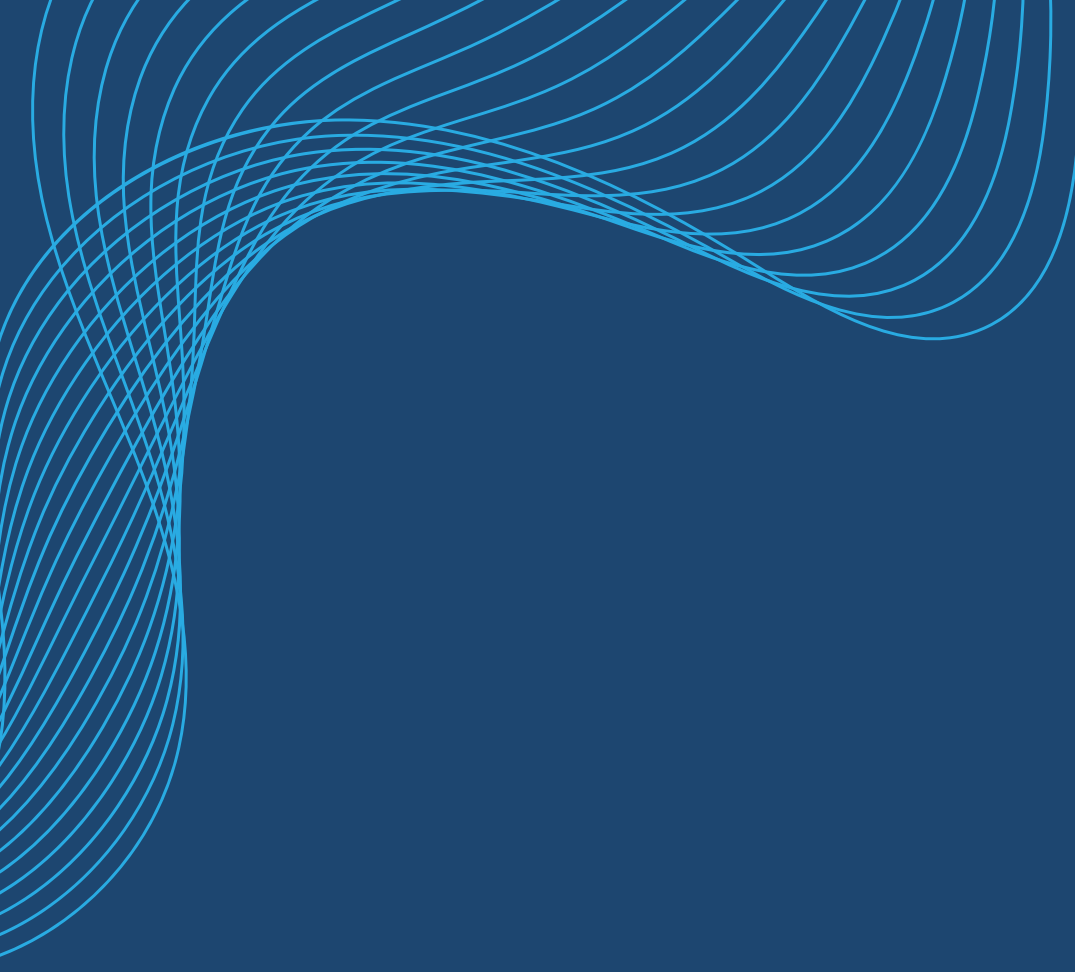
CARL T. BERGSTROM
& JEVIN D. WEST



**WHAT DO YOU THINK IS THE AVERAGE AGE
OF A FIRST KISS?**

KISSING AND SELF-ESTEEM EXAMPLE





HIGHER SELF-ESTEEM



MORE SOCIAL CONFIDENCE



ROMANTIC RELATIONSHIP



KISSING



HIGHER SELF-ESTEEM



CONCLUSION: EXPLORE BEFORE YOU MODEL

Explore first.

Can the data answer the question?

What relationships are present?

What new opportunities appeared?

And underneath:

Sometimes the best model is no model at all.



THANK YOU